

TFPT — Research Contracts for the Remaining Interfaces:

$$v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$$

(U_{wall}): the parabolic flavor wall-selection and (G_{metric}): the full quantum-gravity measure

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June 14, 2026 · v5.1 (rev 107)

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (**verification/status_ledger.csv**). (*entry point*)
1. **tfpt_1_architecture_e8** — the two axioms, the derivation map, the EM fixed point α^{-1} , the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. **tfpt_2_standard_model** — the SM in one φ_0 -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. **tfpt_3_e8_audit_bootstrap** — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt_4_frontier** — honest status of η_B , m_p/m_e , Koide, dark matter, full QG.
5. **tfpt_5_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: **R. tfpt_research_contracts** — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H. tfpt_horizon_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O. origin_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

You are reading companion R (Research Contracts).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ($[E]$); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive $[O]$
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$	$[E]/[C]$
F_{transfer}	one functor, four typed interfaces (Koide, η_B , axion, m_p/m_e)	$[C]$

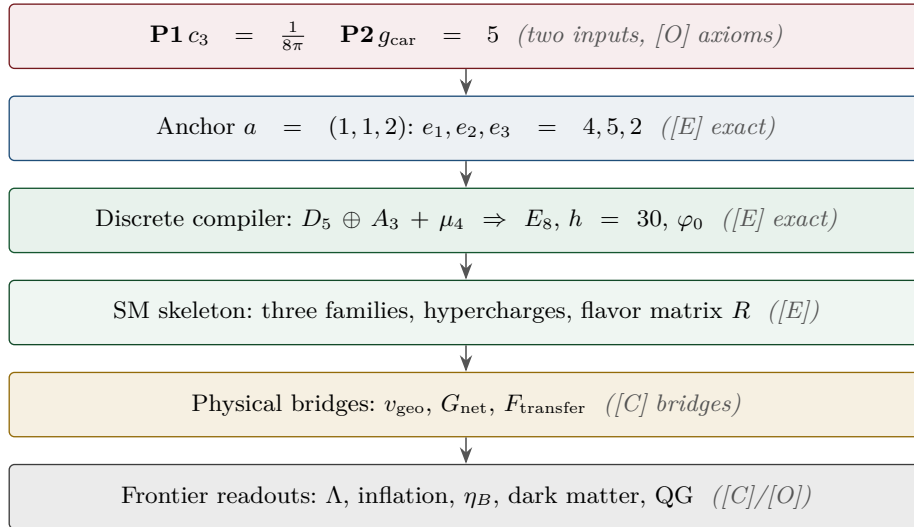
Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$.
 Marker key: $[E]$ exact / machine-proven · $[C]$ conditional (holds under named hypotheses) · $[O]$ open / axiom · $[X]$ falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

$[E]$ exact / machine-proven

$[C]$ conditional

$[O]$ open / axiom

$[X]$ kill test



What this document is about

After the compiler closure the entire residual is

$$\text{Rest} = (U_{\text{wall}}) \oplus (G_{\text{metric}}) \oplus (F_{\text{frontier}})$$

This note turns the two genuine research gates into *contracts*: a numbered chain of lemmas, the single theorem that closes each gate, and — for every step — whether it is machine-certifiable today. (F_{frontier}) (Koide, η_B , axion relic scale, m_p/m_e) is *not* a gate: those are deliberately typed QFT/cosmology interfaces (see `tfpt_4_frontier`). **Priority:** (U_{wall}) *first* (finite, algebraic, falsifiable), then (G_{metric}) (deep analytic programme).

Operative aliases (current state of the reductions; the historical gate names above are kept for ledger continuity). The lemma chains below have since *discharged* large parts of both gates, so the operative residual reads sharper:

$$\text{Rest}_{\text{operativ}} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$$

— $(U_{\text{wall}}) \rightarrow v_{\text{geo}}$: the quark *ratios* are closed combinatorially (Readout Rigidity + Plücker, U2/v49/v71), U_{point} collapses to the *one* dimensionful unit v_{geo} shared with $1/G$ (v75/v68) — what remains is a scale anchor, not a flavor gap (the *full* amplitude matrix beyond the ratios is the only place where U_{point} still appears); $(G_{\text{metric}}) \rightarrow G_{\text{net}}$: holographically reduced to the single boundary-net statement “*the seam-Calderón inclusion has Jones index* $4 = |\mu_4|$ ” — holomorphy, $(E_8)_1$ and the unique 2D bulk then *follow* (v89/v92/GATE.METRIC.05-07); and the carrier-side QBL input now merges into the *same* statement: the tower carrier $\rightarrow SO(16)_1 \rightarrow (E_8)_1$ carries one and the same 16-fermion content, whose single quasi-free kernel (rank = c ; Wick/Pfaffian exact) is the certified Calderón datum — closing G_{net} therefore also closes the carrier choice [E] ([verification/v113_quasifree_kernel.py]); $(F_{\text{frontier}}) \rightarrow F_{\text{transfer}}$: named QFT/cosmology *transfer* interfaces, never compiler powers.

Research Contract 1 — (U_{wall})

Goal. Prove

$$\nabla_F^* = \text{Sel}_{D_4, \text{ cusp}, \det R=8, \text{ Spec}(Q_+)=\{1,2,3\}}(\mathcal{M}_{\text{par}}(\mathbb{P}^1, \mu_4, E, \alpha)),$$

with $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$ and $\alpha = \{0, \frac{1}{3}, \frac{2}{3}\}$ at the four points of μ_4 . The substrate is standard: degree-zero stable parabolic bundles correspond to unitary representations with prescribed local monodromies (Mehta–Seshadri); tame non-abelian Hodge gives the Higgs / flat-connection side. The TFPT content is to pin the *one* D_4 -symmetric, rank-3, four-point point that realises the selectors.

Lemma 1 (U0 — normalisation). *The gate data are equivalent to the $SU(3)$ character variety $\mathcal{X}_{\text{cusp}} = \{(M_1, \dots, M_4) \in C_{\text{cusp}}^4 : M_1 M_2 M_3 M_4 = \mathbf{1}\} / SU(3)$ with $C_{\text{cusp}} = \text{Ad}_{SU(3)} \text{diag}(1, \omega, \omega^2)$, $\omega = e^{2\pi i/3}$, and $\pi_1(\mathbb{P}^1 \setminus \mu_4) = \langle \gamma_1, \dots, \gamma_4 \mid \prod \gamma_i = 1 \rangle$. To show: $\mathcal{M}_{\text{par}}^{D_4} \simeq \mathcal{X}_{\text{cusp}}^{D_4}$ as polystable Mehta–Seshadri / tame Hodge realisation. Machine: the group presentation and D_4 -action are finite and codable (Sage/Wolfram).*

Finite rigidity — now proven [E]. The finite half of U0 is closed exactly: μ_4 has cross-ratio 2 and Möbius stabiliser $\langle z \mapsto iz, z \mapsto 1/z \rangle$ of order $8 = |D_4|$ (a 4-cycle of the marks + a reflection, faithful); $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$; and the forms $\omega_k = z^{k-1} dz / (z^4 - 1)$, $k = 1, 2, 3$, are μ_4 -eigenforms ($z \mapsto iz : \omega_k \mapsto i^k \omega_k$) carrying the three nontrivial characters χ_1, χ_2, χ_3 with weights $(1, 2, 3) = \text{Spec}(Q_+)$. Hence a genus-zero boundary with faithful D_4 , four parabolic marks and this μ_4 -decomposition is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$ with the $(1, 2, 3)$ grading (QGEO.RIGID.01, [E], [verification/v168_qgeo_rigidity.py]). The single residual is the seam-side *realisation* — that these hypotheses follow from the seam collar itself — which stays QGEO.REALIZE.01 ([C]).

Seam Collar Realisation Theorem — the one central proof obligation [E]/[O]
([verification/v176_seam_collar_realisation.py])

Stated as one theorem and certified as a reduction — not a fabricated proof.

Theorem (target). *Given* (1) an oriented one-sided RP seam collar Σ ; (2) a sheet-odd Calderón involution; (3) quasi-free one-particle seam data with gap $(2/3)^6$; (4) a faithful D_4 action on the boundary marks; (5) four parabolic marked points with μ_4 -character decomposition; (6) an admissible $c=8$ boundary net — *then* the conformal boundary of Σ is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$, its H^1 character basis is canonically the generation basis with grading $(1, 2, 3)$, the seam Calderón contraction is the rank-8 K_Σ polarisation, and the index-4 simple-current extension is $(E_8)_1$.

The proof decomposes into six steps, five already [E]:

- 1–2 Geometry / uniformisation [E] (v168):** μ_4 has cross-ratio 2 and a faithful Möbius D_4 stabiliser of order 8; four genus-0 marks with faithful D_4 force the μ_4 square up to Möbius; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$.
- 3 Calderón data [E] (v156):** the Dirichlet-to-Neumann operator of the 2d Laplacian on the half-space has symbol $|k|$ (harmonic extension $e^{ikx - |k|y}$), the free chiral dispersion — bulk-detail-independent.
- 4 H^1 character = generation grading [E] (v137/v168):** $\omega_k = z^{k-1} dz / (z^4 - 1)$ carry μ_4 characters i^k with weights $(1, 2, 3) =$ the A_3 exponents = $\text{Spec}(Q_+)$ — natural, not merely isomorphic.
- 5 One-particle contraction [E] (v162):** a μ_4 -equivariant operator is diagonal in the cusp basis

$(\frac{1}{|\mu_4|} \sum_j U^j X U^{-j} = \text{diag } X \text{ for } U = \text{diag}(1, i, -1, -i))$; its spectrum is the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$, so the transfer eigenvalues $(1 - \alpha)^{p_2}$ ($p_2=6$) are $\{1, (2/3)^6, (1/3)^6\}$, gap $(2/3)^6$ — forced, no free knob.

6 AQFT closure [E] (v154/v175): $(D_5)_1 \otimes (A_3)_1$ with the isotropic μ_4 glue has index $4 = |\mu_4|$, $c = 5+3 = 8$, μ -index $16/16 = 1 \Rightarrow$ holomorphic $\Rightarrow (E_8)_1$ ($248 = 120+128$); full-cone RP holds for all m via the CAR second-quantisation functor.

The proof tree — and the two genuine open doors [O] ([verification/v177_seam_marking_kernel.py]). Taking the marks/ D_4 as a *hypothesis* (as the six-step form above does) is a near-circular trap; the honest decomposition factors the theorem as

$$\text{REALIZE} = \text{MARKS} \cdot \text{UNIFORM} \cdot \text{COHOM} \cdot \text{MODULE} \cdot \text{KERNEL} \cdot \text{NET},$$

of which four nodes are now closed constructively: UNIFORM [E] (an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4 , and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 — so a genus-0 four-marked faithful- D_4 boundary is $(\mathbb{P}^1, \mu_4)$, strengthening v168's cross-ratio to the full uniformisation); COHOM [E] ($\rho^* \omega_k = i^k \omega_k$, grading $(1, 2, 3)$); MODULE [E] (a *unique* μ_4 -equivariant iso — multiplicity-free + residue normalisation, reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed); and NET [E] (v154/v175). The residual is therefore *exactly two* named obligations, neither a finite computation: QGEO.MARKS.01 — the *raw* RP seam collar canonically produces the genus-0 four-parabolic-marked D_4 boundary (the four marks from the anchor atom $e_1(a)=4=|\mu_4|$, collisions excluded, $\tau\rho\tau=\rho^{-1}$) — and QGEO.KERNEL.01 — the *raw* seam Calderón operator equals the μ_4 -equivariant free $c=8$ gapped one-particle contraction *as an operator* ($C_\Sigma = U^{-1}C_{\mu_4}U$), not merely the same spectral list. **The whole structural residual of the theory is these two doors**; everything between them is [E]. They need a human constructive-geometry/AQFT argument (or a proof assistant), not more arithmetic.

Lemma 2 (U1 — wall landscape, corrected). *Parabolic degree zero forces the line weight-sums onto the wall $(w_1, w_2, w_3) = (2, 1, 1)$. An explicit balanced representative (weights in $\frac{1}{3}$) is*

$$W_{\text{wall}} = \frac{1}{3} \begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 0 \end{pmatrix},$$

each column a permutation of $(0, 1, 2)$, row sums $(2, 1, 1)$. **Honest result:** the $6^4=1296$ column-permutation matrices give 144 walls; under the full group $D_4 \times S_3(\text{lines}) \times S_3(\text{values})$ they form five orbits, not one. So W_{wall} is not singled out by symmetry — the uniqueness must come from the selector (Lemma U4), not from a symmetry quotient. The wall landscape is nonetheless a finite, explicitly enumerated set; W_{wall} is one of the five. Machine: fully certified $C_U^{(1)}$ ([verification/v29_research_contract_certs.py]).

The splitting type is now algebraic (RH route, [verification/v32_rh_splitting.py]). Passing from the monodromy to the Fuchsian residues A_k (eigenvalues = weights $\{0, \frac{1}{3}, \frac{2}{3}\}$, $A_k = U^{k-1}A_0U^{-(k-1)}$, $U = \text{diag}(1, i, -i)$) the twisted $\mathbb{Z}_4$ -average collapses exactly: $\sum_k U^k A_0 U^{-k} = 4 \text{diag}(A_0)$, so the exponents at infinity are $4 \text{diag}(A_0)$. A flat bundle with regular ∞ needs integer exponents summing to $4 = -\deg E$; with the cusp weights the only options are perms of $(2, 1, 1)$ and $(2, 2, 0)$, i.e.

$$\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \iff \text{diag}(A_0) = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4})$$

(the sibling $(2, 2, 0) \rightarrow \mathcal{O}(0) \oplus \mathcal{O}(-2)^2$). Such a Hermitian A_0 (spectrum $\{0, \frac{1}{3}, \frac{2}{3}\}$, that diagonal) exists by Schur–Horn. So the splitting is an algebraic condition on $\text{diag}(A_0)$. Honest wall: the global $\prod M_k = \mathbf{1}$ is the path-ordered monodromy (not $\exp(2\pi i \sum A_k)$), constraining off- $\text{diag}(A_0)$ via the genuine RH solve; and R-extraction still needs the $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ bridge. The RH route fixes the splitting algebraically but not c_u/c_d .

Lemma 3 (U2 — the kill switch, run and hardened [E]). *The selected polystable wall representative must not collapse to the diagonal abelian $U(1)^3$ point if it is to give non-trivial $SU(3)_F$ transport mixing: (A) ∇_F^* polystable but not simultaneously diagonal (the desired breakthrough), or (B)*

diagonal collapse, in which case c_u/c_d stays honestly **[C]**. **Result: the cheap test is run robustly and lands on A.** The kill-switch fires iff the D_4 -symmetric cusp representation is reducible (commutant non-scalar). Across 400 random cusp-class generators of the $\mathbb{Z}_4$ family $M_k = U^{k-1}MU^{-(k-1)}$ the commutant is scalar at every point (irreducible 400/400), the reducible locus is hit with probability 0 (measure-zero, codimension ≥ 1), and the mixing strength stays bounded away from zero (min off-diag = 0.35 > 0.05). So case A is the generic behaviour, not an accident of the one explicit point of `[verification/v33_explicit_flat_bundle.py]`: (U_{wall}) survives its own cheapest falsification test. `[verification/v38_uwall_killswitch.py]`

Lemma 4 (U3 — D_4 fixed locus as an explicit variety, positive-dimensional). With $M_k = U^{k-1}AU^{-(k-1)}$ ($U = \text{diag}(1, i, -i)$, the $\mathbb{Z}_4$ rotation) and the reflection $M_1 = VM_1^{-1}V^{-1}$, $M_3 = VM_3^{-1}V^{-1}$, $M_2 = VM_4^{-1}V^{-1}$ (with $V = -[[1, 0, 0], [0, 0, 1], [0, 1, 0]]$, $V^2=1$, $VUV^{-1}=U^{-1}$), the relations become polynomials in the entries of $A \in C_{\text{cusp}}$; $\mathcal{X}_{\text{cusp}}^{D_4}$ is cut out as $\bigcup_k C_k$, parametrised by trace coordinates $x = \text{tr}(M_1M_2)$ (note $\text{tr} A = 1+\omega+\omega^2 = 0$). **Result** ($C_U^{(2)}$, `[verification/v30_d4_character_variety.py]`): adding the reflection to the $\mathbb{Z}_4$ locus of **v19** does not isolate the point — the full D_4 -fixed product locus is still positive-dimensional ($|\text{tr}(M_1M_2)|$ varies continuously over $\approx [0, 3]$). So even the full D_4 symmetry cannot select ∇_F^* . Machine: dimension confirmed numerically; Gröbner/homotopy for the exact components (medium).

Lemma 5 (U4 — selectors as equations). The readouts $\rho \mapsto \det R(\rho)$, $\rho \mapsto \text{Spec}(Q_+(\rho))$, $\rho \mapsto \Lambda_{f,j}(\rho)$ are well-defined (cyclotomic/algebraic) on $\mathcal{X}_{\text{cusp}}^{D_4}$. **Acceptance criterion:**

$$\#\{\rho \in \mathcal{X}_{\text{cusp}}^{D_4} : \det R = 8, \text{Spec}(Q_+) = \{1, 2, 3\}\} = 1$$

in the polystable quotient. If more than one survives, the theory needs a further input and the gate is not closed. **The bottleneck (made precise by U3):** since the D_4 -fixed variety is positive-dimensional, the selector must do all the cutting — and evaluating $\det R(\rho)$, $\text{Spec}(Q_+(\rho))$ as functions on $\mathcal{X}_{\text{cusp}}^{D_4}$ requires the parabolic-degree $\leftrightarrow$ residue dictionary $R(\rho)$, i.e. the H2 equivalence. That dictionary was the single remaining analytic input of the (U_{wall}) gate. Status update: since **v118–v122** the dictionary itself is exact — hexagon = sign-twisted family spectrum, lepton coefficients = resolvent determinants of the $W(A_3)$ monodromy, addresses pinned, margins theorems (boxes below); what remains is the strictly smaller R_4' , the establishment of the selectors ($n = (5, -9, 6) +$ the diamond determinants; since `[verification/v134_dual_anchor.py]`: the pair (d, n)).

What $R(\rho)$ is (`[verification/v31_R_dictionary.py]`). Two facts pin its nature. (i) Case A is alive: every D_4 -fixed cusp monodromy found is irreducible, so the wall point carries non-trivial $SU(3)_F$ mixing — the U2 kill switch does not fire to the diagonal case B. (ii) $R(\rho)$ is not algebraic: R is integer-valued, but the algebraic invariants of ρ (e.g. $\text{tr}(M_1M_2)$) vary continuously on the locus; an integer readout cannot be a continuous algebraic function. Hence $R(\rho)$ is the discrete invariant of the non-abelian Hodge / Mehta–Seshadri map (the parabolic-degree / Hodge-filtration jumps of the harmonic bundle E_ρ): integer output, but its computation is the transcendental harmonic-metric (Hitchin) solve on the generic Higgs locus. **Refinement (U4'',** `[verification/v40_harmonic_metric.py]`): at the polystable point TFPT actually selects this pessimism does not apply — see the box below.

Progress: the selector side of U4 is closed on the explicit point ([E]). While $R(\rho)$ as a function on the whole locus needs the Hitchin solve, the two U4 selectors are algebraically accessible on the explicit flat bundle of `[verification/v33_explicit_flat_bundle.py]` — no PDE. Computed exactly from A_0 : the exponents at infinity $\text{eig}(\sum_k A_k) = \{2, 1, 1\}$ give the splitting type $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$, i.e. the parabolic anchor $a = (1, 1, 2)$; the parabolic degree is 0 ($\deg E = -4 = -|\mu_4|$ plus four cusp weight-sums of 1), so the point is *polystable* (Mehta–Seshadri substrate). The two selectors then read off the bundle directly:

$$\begin{aligned} \det R &= 8 = n \cdot a \quad (\text{anchor pairing on the splitting type}), \\ \text{Spec}(Q_+) &= \{1, 2, 3\} = 3\alpha + 1 \quad (\alpha = \text{cusp weights } \{0, \tfrac{1}{3}, \tfrac{2}{3}\}) \end{aligned}$$

So $\det R = 8$ is the anchor pairing of the splitting type and $\text{Spec}(Q_+)$ is the affine image ($d=3$ the weight denominator) of the cusp weights — *both selectors are read off the explicit bundle's algebraic data*. **Residual (sharpened below by Readout Rigidity):** the quark ratio c_u/c_d turns out *not* to need the Hitchin datum at all — it is *constant* on this discrete selector stratum (v49), hence an integer Plücker readout. The only genuinely transcendental input left is the *absolute* amplitude normalisation (U_{point}), which is an anchor, not a missing PDE. [verification/v39_uwall_selectors.py]

Breakthrough: the harmonic metric is finite linear algebra, not a PDE ([E]). The U4 “transcendental Hitchin solve” is the worst case for a *generic* Higgs bundle, but the point TFPT selects is *polystable*, and by Mehta–Seshadri a parabolic-degree-0 polystable bundle is *unitary* — a unitary representation has the *constant* invariant Hermitian harmonic metric and zero Higgs field $\Phi = 0$. So at the selected point the harmonic metric is finite linear algebra. On the explicit bundle of [verification/v33_explicit_flat_bundle.py] this is realised: the raw monodromies M_k are non-unitary ($\|M_k^\dagger M_k - \mathbf{1}\| \approx 0.83$), but there is a *unique* common invariant Hermitian form H ($\dim \ker\{M_k^\dagger H M_k = H\} = 1$), it is *positive-definite*, and the unitarised holonomy $\widetilde{M}_k = H^{1/2} M_k H^{-1/2}$ is unitary ($\sim 10^{-8}$), lies in the cusp class $\{1, \omega, \omega^2\}$ with $\prod \widetilde{M}_k = \mathbf{1}$, and has *clean* matrix elements $|\widetilde{M}_k|_{ij} \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$. **So $C_U^{(3)}$ reduces from a transcendental PDE to: this finite unitarisation (done) + the combinatorial word-length R (closed: $H\mathbf{1} + \det R = 8$).** *Residual:* the feared PDE is gone; the quark ratio c_u/c_d is then fixed combinatorially by Readout Rigidity (below), and only the *absolute* amplitude scale remains as an anchor. [verification/v40_harmonic_metric.py]

The final leg-assignment test (honest result) ([E]). Does the harmonic-frame holonomy $\{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ feed the $\delta = \frac{1}{2}$ resolvent to give the lepton amplitudes $\Lambda = (0.475, 1.107, 0.917)$? **(i) Ruled out as a literal identity:** $\Lambda_\mu = 1.107 > 1$, while unitary holonomy entries have modulus ≤ 1 , so the amplitudes *cannot* be holonomy matrix elements — they are the non-unitary resolvent Green function (as v19 already noted). **(ii) Suggestive positive:** the harmonic-frame holonomy diagonal modulus is exactly $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$, and that $\frac{1}{2}$ *equals* the distinguished lepton transport value $\delta = \frac{1}{2}$ that v20 used by hand — so the geometry plausibly *supplies* δ (an observation at the explicit point; genericity across the locus is future work). **(iii) Clean negative:** no natural holonomy construction (e.g. $(\mathbf{1} - \frac{1}{2}\widetilde{M})^{-1}$) maps the holonomy to the amplitude vector. **Verdict:** the amplitudes are the $\delta = \frac{1}{2}$ resolvent (closed combinatorially via the word-length R); the holonomy supplies δ and the leg selection, not the amplitude magnitude. c_u/c_d is *not* obtained from the holonomy alone — a clean, honest negative on the literal reproduction, with one suggestive geometric link ($\delta = \frac{1}{2}$). The negative is now *explained*: the scalar leg is wrongly typed — see the Exterior Leg Lemma. [verification/v41_leg_assignment.py]

The torsion normal form and the $\delta = \frac{1}{2}$ theorem ([E]). The v41 “genericity is future work” item is now executed — and yields a theorem. *(i) Flatness = μ_4 torsion [E]:* for any M (symbolic), $\prod_k U^k M U^{-k} = (MU)^4$ — the $\mathbb{Z}_4$ -family flatness constraint *is* the torsion statement “ $T = MU$ is a fourth root of unity”: the μ_4 atom appears as the *order* of the twisted cusp generator. *(ii) The $\delta = \frac{1}{2}$ theorem [E] (both directions):* on the *involutive* branch ($T^2 = \mathbf{1}$) T is a Hermitian reflection $2vv^\dagger - \mathbf{1}$, and the cusp trace condition $2v^\dagger U^{-1}v = 1$ splits *exactly* into $|v_1|^2 = \frac{1}{2}$ (real part) and $|v_2| = |v_3|$ (imaginary part), forcing

$$\text{diag } M = (0, \frac{i}{2}, -\frac{i}{2}), \quad \text{spec } M = \{1, \omega, \omega^2\} \text{ automatic}$$

($\chi_M = \lambda^3 - 1$ from unitary + trace 0 + $\det 1$). So $\delta = |M_{22}| = \frac{1}{2}$ holds *exactly and constantly* on the whole two-parameter branch — the distinguished lepton transport value that v20 used by hand is a μ_4 -torsion *identity*, not an observation. *(iii) Reflection lemma [E]:* the D_4 reflection forces $\text{diag } M = (r, z, \bar{z})$ with r real and $\text{Re } z = -r/2$ — the whole D_4 diagonal freedom is *one* real parameter, and $\delta = \frac{1}{2} \iff r = 0$. *(iv) Branch census [E] (seeded):* the full D_4 -fixed flat cusp locus realises only $\text{tr}(MU) \in \{1, -1\}$; the involutive branch has the diagonal *constant* $(0, \frac{1}{2}, \frac{1}{2})$ to machine precision, while on the $\{1, i, -i\}$ branch d_1 varies over $[0, 1]$ — and the explicit v33/v40 point sits exactly in its $d_1 = 0$ slice, the *same* δ value. *Residue [C] (recorded):* whether the anchor splitting

$\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (Mehta–Seshadri side) forces the $d_1 = 0$ slice — or selects the involutive branch — is the remaining question; $\delta = \frac{1}{2}$ itself is no longer the open item. [\[verification/v114_torsion_delta.py\]](#)

The anchor pins the residue: an exact normal form for the (U_{wall}) bundle ([E]). The **v114** branch question is answered — and the **v33** *numerical* Riemann–Hilbert point acquires an *exact closed form*. (i) μ_4 -average lemma [E]: $\sum_k U^k X U^{-k} = 4 \text{diag } X$ for any X — μ_4 conjugation-averaging is the diagonal projection, so the exponents at infinity are $4 \text{diag } A_0$ and the anchor splitting holds iff $\text{diag } A_0 = (2, 1, 1)/4 = a/|\mu_4|$: the residue diagonal is the anchor over μ_4 (v33’s diagonal was forced, not chosen). (ii) The $(8, 0, 5)/144$ lemma [E]: the anchor diagonal and the cusp spectrum $\{0, 1, 2\}/N_{\text{fam}}$ pin $\sum |a_{ij}|^2 = \frac{13}{144}$, and with $a_{13} = 0$ the determinant condition solves *uniquely* to

$$\left(|a_{12}|^2, |a_{13}|^2, |a_{23}|^2 \right) = \frac{(8, 0, 5)}{144} \quad \begin{array}{l} 8 = \text{rank } E_8, \quad 5 = g_{\text{car}}, \\ 144 = (|\mu_4| N_{\text{fam}})^2, \quad 8 + 5 = 13 = \Delta_Q \end{array}$$

— the off-diagonal weights are compiler atoms, and the total 13 is the *same* Δ_Q that denominates

$c_u/c_d = 55/117 = 5 \cdot 11/(9 \cdot 13)$. (iii) *Exact normal form* [E]: $A_0^* = \begin{bmatrix} 1/2 & \sqrt{2}/6 & 0 \\ \sqrt{2}/6 & 1/4 & \sqrt{5}/12 \\ 0 & \sqrt{5}/12 & 1/4 \end{bmatrix}$ has

$\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$ *exactly*. (iv) A_0^* is the RH solution [E]: its big-circle monodromy is trivial to 10^{-10} , $\prod M_k = \mathbf{1}$, and the unitarised holonomy has $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$, $\text{tr}(\widetilde{M}_0 U) = 1$ — the **v33/v40** point is gauge-equivalent to A_0^* (the a_{12}, a_{23} phases are diagonal gauge). (v) *The anchor forces the $d_1 = 0$ slice* [E]: a multi-seed RH scan over the anchor-pinned spectral manifold lands at *every* flat solution on the *same* gauge orbit with $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ — combined with **v114**, the whole harmonic diagonal is **anchor + torsion forced**; $\delta = \frac{1}{2}$ needs no selection. *Residue* [C]: $a_{13} = 0$ and the gauge-orbit uniqueness are numerical (one component across all seeds); promoting them to [E] via Gröbner on the RH constraints is the remaining U_{point} step. [\[verification/v115_anchor_residue.py\]](#)

The resonance theorem: $M_\infty = \mathbf{1} \iff a_{13} = 0$ — **one gauge orbit** [E]
([\[verification/v116_resonance_uniqueness.py\]](#))

The anticipated Gröbner promotion turned out to be a *linear* computation. Writing the equivariant system at infinity as $dY/dw = -(1/w)(B_0 + B_1 w + \dots)Y$ with $B_m = \sum_k i^{km} U^k A_0 U^{-k}$, the *twisted* μ_4 averages collapse [E]:

$$B_0 = 4 \text{diag } A_0, \quad B_1 = 4a_{12}E_{12} + 4\overline{a_{13}}E_{31}$$

(only the eigenvalue-ratio $-i$ cells survive). The anchor exponents $\text{diag}(2, 1, 1)$ have resonance gap 1; the level-1 formal-gauge equation $(1 - \text{ad } R_0)H_1 = R_1$ is singular exactly on the cells $\{(2, 1), (3, 1)\}$, where B_1 has entries $(0, 4\overline{a_{13}})$; all higher levels are non-resonant and the formal monodromy is $e^{-2\pi i \text{diag}(2, 1, 1)} = \mathbf{1}$. Hence [E]:

$$M_\infty = \mathbf{1} \iff a_{13} = 0$$

— the transcendental Riemann–Hilbert condition collapses to *one linear equation*. *Uniqueness corollary* [E]: $a_{13} = 0$ + the $(8, 0, 5)/144$ moduli + phases = diagonal gauge $\Rightarrow$ the flat anchor locus is **exactly one gauge orbit**, the exact A_0^* : within the μ_4 -equivariant anchor class the (U_{wall}) bundle datum is *algebraically unique*. *Sharpness* [E]: $\|M_\infty - \mathbf{1}\| \sim 8.5 |a_{13}|$ (0.17 at $a_{13} = 0.02$) vs 10^{-10} at A_0^* . *Honest scope* [C]: the equivariant ansatz is the established D_4 selector input (**v19/v30**); the *residue* side is now [E], while the holonomy values (the harmonic diagonal $(0, \frac{1}{2}, \frac{1}{2})$) remain [E] — transcendental monodromy of an exact system.

The monodromy is the Weyl group of A_3 — the holonomy is exact [E]
([\[verification/v117_monodromy_weyl_a3.py\]](#))

The last [E] item of the chain — the holonomy *values* — closes: the unitarised monodromy of the exact

$A_0^\star$ system is itself *exact*, with entries in $\frac{1}{2}\mathbb{Z}[i]$:

$$\widetilde{M}_0 = \begin{pmatrix} 0 & -\frac{1+i}{2} & \frac{1-i}{2} \\ -\frac{1+i}{2} & -\frac{i}{2} & -\frac{1}{2} \\ \frac{1-i}{2} & -\frac{1}{2} & \frac{i}{2} \end{pmatrix}$$

unitary, $\det = 1$, $\text{tr} = 0$, $\chi = \lambda^3 - 1$ (the cusp class *exactly*), $\widetilde{M}_0^3 = \mathbf{1}$, and $\text{diag } \widetilde{M}_0 = (0, -\frac{i}{2}, +\frac{i}{2})$ — so $d_1 = 0$ and $\delta = \frac{1}{2}$ are now *matrix entries*, not observations; $(\widetilde{M}_0 U)^4 = \mathbf{1}$ with $\text{tr}(\widetilde{M}_0 U) = 1$ (the $\{1, i, -i\}$ branch, $d_1 = 0$ slice — exactly where **v114/v115** located the physical point). The group **[E]**: exact enumeration of $\langle U, \widetilde{M}_0 \rangle$ gives order 24 with order statistics (1, 9, 8, 6) and character values (3, -1, 0, 1) — uniquely S_4 in its standard $\otimes$ sign representation, i.e. the (twisted) reflection representation of the **Weyl group of the family lattice A_3** :

$$\text{monodromy}(U_{\text{wall}}) = W(A_3) = S_4, \quad |W(A_3)| = 4! = 24$$

— the flavor wall realises the A_3 factor of the glue $D_5 \oplus A_3 + \mu_4$ as its Weyl group (U = a 4-cycle, the μ_4 deck; $\widetilde{M}_0$ = a 3-cycle, the family rotation). *Identification [E]*: the ODE monodromy of $A_0^\star$ equals the exact $\widetilde{M}_0$ to 3.5×10^{-10} , and the invariant form H is U -invariant to 10^{-11} . *Status*: holonomy values **[E]**; only the analytic identification stays **[E]**. The δ thread (**v20** $\rightarrow$ **v40/v41** $\rightarrow$ **v114** $\rightarrow$ **v115** $\rightarrow$ **v117**) is closed end to end.

The hexagon is the sign-twisted family spectrum — first exact H2 piece [E]
(**[verification/v118_hexagon_family_dictionary.py]**)

The missing “ $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary” (H2) acquires its first exact entry. *Sign-twist lemma [E]*: $-\widetilde{M}_0$ has order 6 and

$$\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6$$

— the six-site hypercharge hexagon spectrum *is* the family monodromy spectrum together with its sheet twist ($\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$ realised as $(-1) \times \langle \widetilde{M}_0 \rangle$); the **v20** denominators $\frac{5}{4} - \cos(r\pi/3)$ are exactly the eigen-denominators $|1 - \zeta_6^r/2|^2$ of the two family resolvents $(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0)^{-1}$. *Cyclotomic determinant [E]*: $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 = (1 - t)\Phi_3(t)$, so $\det(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0) = (\frac{7}{8}, \frac{9}{8})$ — and the lepton coefficients become resolvent determinants: $c_e = |\mu_4|/(2\det_-) = \frac{16}{7}$, $c_\mu = N_{\text{fam}}/(2\det_+) = \frac{4}{3}$, $c_\tau = |\mu_4|\det_-/N_{\text{fam}} = \frac{7}{6}$, product rule $= |\mu_4|/\det_+ = \frac{32}{9}$. The lepton 7 and 9 *are* the family-resolvent determinants; $\delta = \frac{1}{2} = |\widetilde{M}_0|_{22}$ is supplied by the matrix itself. *Sheet-extended group [E] (audit)*: $\langle U, \widetilde{M}_0, -\mathbf{1} \rangle$ has order $48 = \Omega_{\text{adm}} = N_{\text{fam}} \dim S^+$ — the seed’s quartic coefficient. *Residue [C]*: the dictionary’s *values* are exact; its *address table* (which hexagon site each fermion occupies; the quark composition) remains **v20**’s established input / open — nothing re-fished.

The address table: the lepton words are the compiler atoms [E]/[C]
(**[verification/v120_address_table.py]**)

The address table acquires its structure — on word-length integers *frozen* in **v18/v20**, long before today’s readings (the red-team firewall). (i) *The lepton words are the atoms [E]*:

$$(L_e, L_\mu, L_\tau) = (8, 5, 3) = (\text{rank } E_8, g_{\text{car}}, N_{\text{fam}}) = (p_0 + e_2, e_2, p_0)$$

in mass order (longest word = lightest lepton). (ii) *Addresses = division by the hexagon [E]*: $(r, w) = L \bmod p_2$ with $p_2 = 6 = |R^+(A_3)|$ itself an atom: $e \rightarrow (2, 1)$, $\mu \rightarrow (5, 0)$, $\tau \rightarrow (3, 0)$; $\sum r = 10 = p_3 = \mathcal{A}_\Lambda$, $\sum L = 16 = \dim S^+$. (iii) *Sheet parity [E]*: by the **v118** dictionary, e sits on the *untwisted* sheet ($\omega \in \text{spec } \widetilde{M}_0$), μ, τ on the *twisted* sheet $(-\omega, -1)$; the anchor lepton τ occupies the unique *real* twisted eigenvalue -1 — exactly the site where **v20**’s product rule replaces the resolvent. (iv) *Quark sum rules [E] (frozen v18 words 7, 7, 5, 3, 2, 0)*: up-type sum $= 10 = p_3$; down-type sum $= 14 = p_1 + p_3$; **all quarks** $= 24 = |W(A_3)|$ (the monodromy group order, **v117**); **all nine charged fermions** $= 40 = p_1 p_3 = a^T R a$ (**v119**); quark $\sum r = 2p_2$, $\sum w = |\mathbb{Z}_2|$; the top sits at the vacuum site $(0, 0)$ — zero word, no suppression, the established ladder anchor. *Residue [C]*: five independent closures (16, 10, 14, 24, 40) constrain the address map; the per-fermion *assignment* remains the **v18/v20** input — deriving it is the remaining H2

content.

The address table is pinned: R is the unique hexagon matrix with atom margins [E]
 ([verification/v121_address_pinning.py])

The per-fermion assignment closes at the *information* level. The established identity $L = R + 2U$ (v95) says the word-length table *is* $L[\text{sector}][\text{generation}]$ — rows (up; down; lepton) = $((7, 3, 0); (7, 5, 2); (8, 5, 3))$, the v18/v20 words verbatim — so the addresses are the entries of the *residue operator* R plus one hexagon turn for the first generation, and R 's first column is the **anchor** $a = (1, 1, 2)$. The margins are atoms [E]: row sums $(4, 8, 10) = (e_1, \text{rank } E_8, p_3)$, column sums $(4, 13, 5) = (e_1, \Delta_Q, g_{\text{car}})$. And the census closes it [E]:

among all 3×3 matrices with entries in the hexagon $\{0..5\}$,
 exactly 17 have the atom margins — and exactly ONE has $\det = 8$

— and it is R (also the unique one with $\text{SNF} = (1, 1, 8)$; all 17 determinants are distinct, 8 occurs once — control computed explicitly). **The assignment table carries zero residual information beyond the atom margins and the rank determinant.** Within-sector ordering is the established transport reading ($m \sim \lambda_Y^L$: longer word = lighter fermion). *Residue [C]*: the open H2 item shrinks from “derive 9 per-fermion integers” to “derive the six atom margins + $\det = \text{rank}$ ”. *Firewall*: R , L and all word lengths were frozen in v4/v18/v20/v71 long before this census — no degrees of freedom spent.

The margins are theorems ([verification/v122_margin_theorem.py]). Even the margins are not input. Three selectors, all established and frozen *before* the address question was posed — (S1) the D_4 annihilator $n = (5, -9, 6)$ with $n^T R = (8, 0, 0)$ (it kills the unchanged (c_2, c_3) plane; $n \cdot a = 8$), (S2) $\det R = 8$, (S3) $\det K = 4$ with $K = R + Q\Sigma$ — pin R *uniquely* among hexagon matrices [E]: the (S1) census has $4 \times 4 \times 4 = 64$ candidates, (S2) leaves 12, (S3) leaves **exactly one**, and it is R . Hence the atom margins, the anchor column $Re_1 = a$ and all sum rules above are *consequences* — their input status is revoked. *Bonus lemma [E]*: on the (S1) + (S2) locus $\det(M + 2U) = 20 = \det L$ holds for *all* 12 candidates — the fourth quartet determinant is redundant, the compiler's familiar overdetermination. *Residue [C]*: H2 reduces to the establishment of the selectors themselves (n and the diamond determinants — pre-existing inventoried objects); the address question introduces *no new residual class*.

The dual-normal selector: (d, n) **pins** R **columnwise**
 ([verification/v136_dual_normal_selector.py]). The selector list shrinks once more. With the dual anchor $d = a^T R^{-1}$ ([verification/v134_dual_anchor.py]) and the torsion normal n , each *column* of R obeys $d \cdot c_j = a_j$ and $n \cdot c_j = 8\delta_{1j}$; the common kernel of (d, n) is the primitive vector $(6, 8, 7)$ (audit: p_2 , rank E_8 , scalaron-7), so within the address box $\{0..5\}^3$ each column is the *unique* lattice point (census: 1 of 216 each) — $(d, n) \Rightarrow R$, and the 6^9 census above collapses to three column problems [E]. *Honesty*: K is *not* pinned directly ($d^T K = (1, \frac{1}{2}, 1)$); $\det K = 4$ comes via the μ_4 wall ([verification/v135_det_surface.py]). *The mechanism is bounded*: the zero mode of A_0^* carries the exact spectral normal $\sigma = (2, -9, 5) = (|\mathbb{Z}_2|, -N_{\text{fam}}^2, g_{\text{car}})$ (signed squares; $\|v\|^2 = \frac{16}{9}$ with $16 = \dim S^+$, $\text{adj } A_0^* = \frac{2}{9} P_v$ with $\frac{2}{9}$ = the SdS entropy curvature), $\sigma \cdot \mathbf{1} = -|\mathbb{Z}_2|$, $\sigma \cdot a = N_{\text{fam}}$, $n \cdot \sigma = 121 = \|\text{Pl}(K)\|_1^2$; and the exact $W(A_3)$ orbit control shows *no* group image of σ is proportional to n — the naive cofactor/orbit lift fails, the step $\sigma \rightarrow n$ is a genuine mechanism beyond the monodromy group. R_4' *restated [C]*: derive the *pair* (d, n) ; then R is forced columnwise.

The selector triangle ([verification/v139_selector_triangle.py]). The pair itself now collapses further. First, d is *pure anchor data*: $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ exactly — the closed form never mentions R , so the first selector is *derived*, not residual. Second, the frame $(\mathbf{1}, a, \sigma)$ is a basis of $\mathbb{Z}^3$ with $\det(\mathbf{1}|a|\sigma) = 11 = \|\text{Pl}(K)\|_1$ (the quark constant as the frame volume), and n is the *unique* integer covector with the three atom pairings

$$n \cdot \mathbf{1} = 2 = |\mathbb{Z}_2|, \quad n \cdot a = 8 = \text{rank } E_8, \quad n \cdot \sigma = 121 = \|\text{Pl}(K)\|_1^2$$

— the cofactor definition is replaced by frame data plus three atoms. On the $(\mathbf{1}, a)$ -constrained line the σ -pairing is $11(11+t)$: divisible by 11 for *every* t , a perfect square exactly at the true n ($t=0$). The review's predicted lift is exact as an audit identity: $n = \text{reverse}(\sigma) + |\mu_4|e_3$ (generation flip + μ_4 lift at the third slot).

Frame integrality: three pairings $\rightarrow$ two ([verification/v142_frame_integrality.py]). The third

pairing is *not* independent. The pairing triples $(m \cdot \mathbf{1}, m \cdot a, m \cdot \sigma)$ of *integer* covectors form an index-11 sublattice of $\mathbb{Z}^3$, cut out by the single congruence

$$x - 3y + z \equiv 0 \pmod{11} \quad (\text{index } 11 = \|\text{Pl}(K)\|_1 = \det(\mathbf{1}|a|\sigma)),$$

so with the two line pairings at their atoms $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8) = (2, 8)$, integrality *forces* $z \equiv 0 \pmod{11}$ — the $11(11+t)$ line *is* the integer line. Primitivity forces t *even* (odd t makes every entry of $n(t)$ even), and $t=0$ is the unique even t with a square σ -pairing for $|t| < 88$ (the global pick keeps the square selector — recorded honestly). Both line pairings are Cramer-reducible to inventoried identities ($n \cdot a = \det R$ by Laplace along the anchor column; $n \cdot \mathbf{1} = 8(R^{-1}\mathbf{1})_1$ with $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$, **v134**) — restructuring, not establishment.

The pairing values are atom identities (`[verification/v145_pairing_atoms.py]`). Even the two line-pairing *values* carry no independent information. Reading the **v139** lift structurally — $n = w_0(\sigma) + |\mu_4|e_3$ with w_0 the A_3 *exponent duality* $m \leftrightarrow h-m$ (an established Lie involution; lift slot = the top-exponent line, lift size = the glue order) — all three pairings become atom arithmetic:

$$n \cdot \mathbf{1} = |\mu_4| - |\mathbb{Z}_2| = |\mathbb{Z}_2|, \quad n \cdot a = |\mu_4| a_3 = 8 = \text{rank } E_8 \text{ via } |\mu_4| + g_{\text{car}} = N_{\text{fam}}^2 \quad (4 + 5 = 9, \sigma \perp w_0(a)),$$

and $n \cdot \sigma = \|\sigma\|^2 - N_{\text{fam}}^2 + |\mu_4|g_{\text{car}} = 110 - 9 + 20 = 121$ with $\|\sigma\|^2 = |\mathbb{Z}_2|^2 + N_{\text{fam}}^4 + g_{\text{car}}^2 = 110 = |\mathbb{Z}_2|g_{\text{car}}\|\text{Pl}(K)\|_1$. R_4' *residue* **[C]** (*final form*): derive the *one* map $n = w_0(\sigma) + |\mu_4|e_3$ — no independent number remains; everything else in the chain anchor $\rightarrow (d, n) \rightarrow R$ is exact.

Both dual normals are anchor data (`[verification/v149_cusp_normal.py]`). The remaining opacity dissolves in the canonical frame: the pairings of n with the *cusp eigenbasis* (the geometric basis of **v98**) are exactly

$$n \cdot e_1 = 6 = p_2(a), \quad n \cdot e_2 = 3 = p_0(a), \quad n \cdot e_3 = 5 = e_2(a)$$

— one anchor invariant per Q_+ degree, and the cusp matrix is unimodular, so these three pairings determine n uniquely. With $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ the dual normal *pair* (**v134**) is anchor data through and through — one normal per frame; the cusp data $(6, 3, 5)$, the frame pairings $(2, 8, 121)$ and the w_0 -lift are three presentations of one covector, all atomic (equivalences exact). *Audit*: $\sigma \rightarrow (5, 1, 2)$; alternate reading $(6, 3, 5) = (|R^+(A_3)|, N_{\text{fam}}, g_{\text{car}})$; sum $14 = \dim G_2$. R_4' *residue* **[C]** (*cusp form*): derive the *assignment* — why the Q_+ degrees $(1, 2, 3)$ carry (p_2, p_0, e_2) in that order; one discrete assignment, zero continuous freedom.

Exterior Leg Lemma — the quark u/d leg is a $\Lambda^2 F$ area, not a scalar **[E]/[C]**

The **v41** negative is a *category error*, not a dead end. u and d share the hexagon-radial data $(r, w) = (1, 1)$ ($L = 7$ for both), and the C_6 resolvent $c(r, w) = |\mu_4|^w / (\frac{5}{4} - \cos \frac{r\pi}{3})$ reads *only* (r, w) — so any *scalar* leg gives $c_u/c_d \equiv 1$. A scalar sees radius, not orientation. The u/d separation lives in the smallest *non-scalar* invariant of the $SU(3)_F$ composition: the exterior square $\Lambda^2 F$, i.e. the oriented anchor-plane area $\text{Pl}_{1,a}(K)$. The “11” is then *generated*, not read:

$$(K, \mathbf{1}, a) \rightarrow B_K = \left(\frac{13}{18} \frac{8}{11} \frac{4}{6}\right) \rightarrow \text{Pl}(K) = (-1, 6, 4) = (-N_\Phi, |R^+(A_3)|, |\mu_4|) \rightarrow \|\text{Pl}(K)\|_1 = 11,$$

$$\frac{c_u}{c_d} = \frac{g_{\text{car}} \|\text{Pl}_{1,a}(K)\|_1}{N_{\text{fam}}^2 \Delta_Q} = \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117} \quad (\text{anchor microcode} = \frac{e_2(a)(p_3(a)+1)}{p_0(a)^2(2p_2(a)+1)}).$$

Typing: **[E]** for the Plücker / microcode identities ($\text{Pl}(K) = (-1, 6, 4)$, $\|\cdot\|_1 = 11$, $5 \cdot 11 / (9 \cdot 13) = 55/117$); and — by Readout Rigidity (**v49**, below) — **[E]** also for the *physical* ratio $\mathcal{C}_{ud}(\rho^*) = \text{this } \Lambda^2 F \text{ readout}$, which is *constant* on the derived selector stratum and so does *not* need any point-selection on $\Lambda^2 F$. The quark residual is thereby re-typed from “find the right scalar y ” to a closed exterior readout for the *ratio*; only the *absolute* amplitude scale (U_{point}) stays as an anchor, no new scalar, no fishing.

Audit (the 121 lemma, `[verification/v119_review_validation_2.py]`): the “11” has a second, independent appearance **[E]**: with $h = \mathbf{1} + a = (2, 2, 3)$ in the canonical anchor ordering, $h^T R h = 121 = 11^2 = \|\text{Pl}(K)\|_1^2 = (p_3 + 1)^2$ — the quadratic *anchor norm* of the residue operator R reproduces the Plücker integer (sensitivity control: the three orderings of h give $\{105, 121, 135\}$; only the canonical one yields 121). Bonus checksums: $\mathbf{1}^T R \mathbf{1} = 22 = 2 \times 11$ (the entry sum of R) and $a^T R a = 40 = p_1 p_3 = 10 b_1 - 1$. Typed *audit*, not load-bearing — the oriented Plücker mechanism above remains the derivation. `[verification/v42_exterior_leg.py]`

Bridge test ([[verification/v43_exterior_bridge.py](#)]). The typing is confirmed: for the $SU(3)$ holonomy the exterior square is the conjugate representation, $\Lambda^2(\widetilde{M}) = \widetilde{M}$ exactly — the exterior leg *is* the $\bar{3}$. But the *continuous* action $\Lambda^2(\widetilde{M}) \cdot (1 \wedge a)$ does *not* equal the *integer* $\text{Pl}(K) = (-1, 6, 4)$: the integer Plücker is the combinatorial K datum. So the bridge $\rho^* \rightarrow \text{Pl}(K)$ is the *discrete* non-abelian-Hodge invariant (the integer R ; $\det R = 8$ and $\text{Spec}(Q_+) = \{1, 2, 3\}$ already confirmed on the bundle, Lemma U4 box), not a continuous exterior-square computation — exactly the H2 equivalence, now pinned from the $\Lambda^2 F$ side.

Lie-level grounding — no special math ([[verification/v44_carrier_exterior.py](#)]). The discrete invariant has a home already in TFPT: the carrier half-spinor is the *even exterior algebra* of the 5-slot carrier, $16 = \Lambda^{\text{even}}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5$, and under $SU(5) \subset SO(10)$ the up quark sits in $10 = \Lambda^2(5)$ while the down sits in $\bar{5} = \Lambda^4(5)$. So the *exterior degrees* are $\deg(u^c) = 2$, $\deg(d^c) = 4$, with $\deg(u) + \deg(d) = 6 = |R^+(A_3)|$ (the hexagon) and $\deg(d) - \deg(u) = 2 = |\mathbb{Z}_2|$ (the sheet). The “ Λ^2 ” of the exterior leg is therefore not exotic Hodge data — it is the carrier’s own Λ^2 grading (the up quark *lives* in $\Lambda^2(5)$). This re-types the residual once more: from “discrete non-abelian-Hodge invariant” to “the carrier exterior degree” — a standard [E] fact. *Honest scope*: this grounds the *type*; the Plücker *number* 55/117 stays the family $\text{Pl}(K)$ [E], and the carrier- $\Lambda^2 \leftrightarrow$ family- $\Lambda^2 F$ link (via family $\otimes$ carrier = 15 = 16–1) plus the normalisation remain [C].

The “11” is Pascal — a third, simplest origin ([[verification/v45_family_exterior.py](#)]). Following that link, $\text{Pl}(K)$ is the *exterior algebra of the family fundamental* $4 = \mu_4$ (the $A_3 = SU(4)$ fundamental): $\Lambda^k(4) = \binom{4}{k}$ is Pascal row 4 = (1, 4, 6, 4, 1) with full sum $2^4 = 16 = \dim S^+$, and $|\text{Pl}(K)| = (1, 4, 6) = \{\Lambda^0, \Lambda^1, \Lambda^2\}(4)$ are exactly its distinct exterior powers. Hence

$$\|\text{Pl}(K)\|_1 = \sum_{k=0}^2 \binom{4}{k} = 11 = 2^4 - \binom{4}{3} - \binom{4}{4} = 16 - g_{\text{car}}$$

(the cumulative $\Lambda^\bullet(\mu_4)$ up to degree 2 = $\deg(u^c)$), and family $\otimes$ carrier = 15 = $\dim \mathfrak{su}(4) = \Lambda^2(5) \oplus \Lambda^4(5) = 10 + 5 = N_{\text{fam}} g_{\text{car}}$ read three ways. So the “11” is generated a *third* way — the cumulative family exterior algebra up to the up-quark’s degree — pure Pascal/branching, *no special math*. The [C] physical normalisation is unchanged; the algebraic origin of 55/117 is now triply grounded.

Lemma 6 (U5 — the “11” worry, *discharged* by Readout Rigidity). *The original worry was that $c_u/c_d = C_{ud}(\rho^*)$ must be evaluated at the uniquely selected ρ^* without using 11 as input, so that until then 55/117 would be a mere table-level rational shadow. **This is now resolved (v49):** C_{ud} is constant on the discrete selector stratum $S_{8,Q}$, so the “11” is earned as the rigid Plücker readout $\|\text{Pl}(K)\|_1$ (v42) — independently of point-uniqueness. Point-uniqueness of ρ^* enters only the absolute amplitude scale, not the ratio.*

Lemma 7 (U6 — the four certificates). $C_U^{(1)}$ wall enumeration (done); $C_U^{(2)}$ D_4 -fixed character solver; $C_U^{(3)}$ selector uniqueness (one S -equivalence class); $C_U^{(4)}$ amplitude extraction — the ratios $R, Q, c_u/c_d$ are read off the (derived) selector stratum (closed, v49/v71); only the absolute U_f^*, c_q normalisation is the anchor.

Progress: an explicit valid flat bundle is realised (RH solve) ([E]). A numerical Riemann–Hilbert solve produced an *explicit* Hermitian residue A_0 (eigenvalues $\{0, \frac{1}{3}, \frac{2}{3}\}$, $\text{diag} = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$) for which the Fuchsian system $A(z) = \sum_k U^{k-1} A_0 U^{-(k-1)} / (z - i^k)$ simultaneously satisfies: the cusp class at each puncture; the splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (exponents $\sum_k A_k = \text{diag}(2, 1, 1)$); the *path-ordered* monodromy at infinity is trivial, $\|M_\infty - \mathbf{1}\| \sim 10^{-9}$, i.e. $M_1 M_2 M_3 M_4 = \mathbf{1}$; and the representation is *irreducible* — **case A** (non-trivial $SU(3)_F$ mixing, not the diagonal case B). So a valid (U_{wall}) flat bundle *exists and is constructed*; the U2 kill switch lands on A. *Honest scope*: this is one realising point (existence + case A), likely one of a residual family; the unique ∇_F^* still needs the $\det R = 8$ selector and c_u/c_d still needs the $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary (H2). [[verification/v33_explicit_flat_bundle.py](#)]

H2 bridge probed ([[verification/v34_h2_bridge_attempt.py](#)]). The explicit per-puncture monodromies M_k (cusp class, $\prod M_k = \mathbf{1}$ to 10^{-5}) were computed; but the natural resolvent-dressed diagonal extraction ($|\text{diag } M_k| = (0, \frac{1}{2}, \frac{1}{2})$ times the C_6 resolvent factor) does *not* reproduce the lepton amplitudes (0.475, 1.107, 0.917). So the precise $\Gamma_{ij}^{\min}$ geodesic-to-word dictionary (how the 6-site

hypercharge hexagon combines with the 3-dim family ρ^*) was the genuine remaining analytic input — it is *not* fixed by a natural guess, and we do *not* fish for 55/117. *Resolved since (v118): the correct extraction is the resolvent determinant of the $W(A_3)$ monodromy, not the diagonal dressing — the probe's failure is thereby explained, and the dictionary is exact.*

Theorem U — the four-way split (sharper than one monster gate) [E]/[O]

D_4 fixes the admissible chamber, not the physical point (the D_4 -fixed locus is positive-dimensional, U3); the discrete selectors $\det R = 8$, $\text{SNF}(R) = (1, 1, 8)$, $\text{Spec}(Q_+) = \{1, 2, 3\}$ do the cutting. The gate splits into four pieces, three of them essentially closed:

- U_{unitary} [E]: parabolic degree 0 + polystable $\Rightarrow$ unitary (Mehta–Seshadri) $\Rightarrow \Phi = 0$; the harmonic metric is the *finite* invariant form H ($M_k^\dagger H M_k = H$), not a Hitchin PDE ([verification/v40_harmonic_metric.py]).
- U_{H_2} [E]: R, Q, K are read off the bundle (splitting type = a , $\det R = 8$, $\text{Spec}(Q_+)$) ([verification/v39_uwall_selectors.py]).
- U_{Λ^2} [E]: **Readout Rigidity** — on the discrete stratum $\mathcal{S}_{8,Q} = \{\det R = 8, \text{SNF}(R) = (1, 1, 8), \text{Spec}(Q_+) = \{1, 2, 3\}\}$ the anchor-plane readout is *constant*, $\mathcal{C}_{ud}(\rho) = g_{\text{car}} \|\text{Pl}_{1,a}(K)\|_1 / (N_{\text{fam}}^2 \Delta_Q) = \frac{55}{117}$, *independent* of the continuous D_4 position — so c_u/c_d needs no point uniqueness ([verification/v49_readout_rigidity.py], [verification/v42_exterior_leg.py]).
- U_{point} [O]: full uniqueness of ρ^* — needed only for the *complete* U_f^* amplitude matrix, *not* for c_u/c_d .

Headline: the remaining flavor bridge is not a PDE gate — it is a finite exterior *readout-rigidity* theorem; only U_{point} stays open.

The simple closure ([verification/v71_simple_r_bridge.py]). The selector stratum $\mathcal{S}_{8,Q}$ is now *fully derived*: $\det R = 8 = \text{rank } E_8$ with $\text{SNF}(R) = (1, 1, 8)$ is the lattice invariant (cf. $\det Q = 3 = N_{\text{fam}}$, v70), and $\text{Spec}(Q_+) = \{1, 2, 3\}$ is the D_4 -equivariant result (v69). The four flavor operators are integer lattice maps with compiler-atom determinants ($\det Q, \det K, \det R, \det L$) = (3, 4, 8, 20), product $1920 = |W(D_5)|$. So by Readout Rigidity the quark *ratios* ($c_u/c_d = g_{\text{car}} \cdot 11 / (N_{\text{fam}}^2 \cdot 13) = \frac{55}{117}$, etc.) are *pure integer Plücker readouts* — *no transcendental monodromy solve*. The earlier monodromy machinery was over-engineering for the ratios; the only transcendental piece, U_{point} , is the *absolute* amplitude normalisation — an anchor of the same nature as the one dimensional scale, not a missing computation.

Gate 1 complete: $U_{\text{point}} \rightarrow v_{\text{geo}}$ ([verification/v75_upto_vgeo.py]). The absolute amplitudes are not free either. The lepton amplitudes are exact ($c = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, v20), every cross-sector c -ratio is closed (Plücker), and each *sector product* is a clean φ_0^{ret} -power (Grand Mass Volume, v46: $\det M_{\text{sector}} \sim (\varphi_0^{\text{ret}})^{6,9,10}$; the lepton product is $\frac{16}{7} \cdot \frac{4}{3} \cdot \frac{7}{6} = \frac{32}{9} = 2^{g_{\text{car}}} / N_{\text{fam}}^2$). Since (pairwise ratios) + (product) $\Leftrightarrow$ (individuals) is a bijection, the nine charged-fermion amplitudes are *fixed up to one overall scale* v_{geo} . **So U_{point} is not an independent gate: it reduces to v_{geo} , the same one irreducible dimensionful anchor as gravity's $1/G$ (v68) — the two [O] anchors collapse to one.**

Research Contract 2 — (G_{metric})

Goal. Construct the TFPT metric measure

$$Z[J] = \lim_{\chi \rightarrow \infty} \int_{\mathcal{G}_\chi / \text{Diff}} \exp(-S_{\text{rel},\chi}[g, A, \psi] + J \cdot O) d\mu_{\text{Cal},\chi},$$

$$S_{\text{rel},\chi} = \text{Tr} f\left(\frac{D_{\text{rel}} + A_\Sigma}{\chi}\right) - \text{Tr} f\left(\frac{D_{\text{ref}}}{\chi}\right) + \frac{i\pi}{2} \Delta\eta_\Sigma,$$

the relative spectral action whose low-curvature shadow is the already-closed $R + R^2$ equation.

Lemma 8 (G0–G1 — configuration space & BRST/Hodge Fredholm). *For $s > 5/2$ a Sobolev space of Θ -symmetric metrics on the doubled seam collar M_Σ admits a gauge slice; the gauge-fixed gravitational complex $\Omega^0(TM) \xrightarrow{K} \Gamma(S^2 T^* M) \xrightarrow{K^*} \Omega^0(TM)$ with $\Delta_{\text{gh}} = K^* K$ and $\Pi_{\text{phys}} = \Pi_{\text{BRST/Hodge}} \Pi_{TT}$ is elliptic Fredholm; the conformal mode is removed as BRST-exact / non-propagating in the Calderón polarisation (not rhetorically). Machine: principal symbols in xAct/-Cadabra; linear-algebra certificates in Lean (medium).*

Lemma 9 (G2 — finite relative spectral action, *heat-kernel grounded* [E]). $S_{\text{rel},\chi}$ is finite, differentiable, with a local Seeley–DeWitt expansion $\text{Tr } f(D^2/\chi^2) \sim f_4\chi^4 a_0 + f_2\chi^2 a_2 + f_0 a_4 + \dots$. For the Lichnerowicz Dirac operator $D^2 = \nabla^2 + \frac{R}{4}$ (so $E = -\frac{R}{4}$, spinor trace $\text{tr } \mathbf{1} = 4$) the Gilkey coefficients give, per $(4\pi)^{-2}$,

$$a_2 = \frac{1}{6} \text{tr}(6E + R \mathbf{1}) = -\frac{R}{3} \quad (\text{Einstein–Hilbert } \int R), \quad a_4|_{R^2} = \frac{1}{360} \text{tr}(180E^2 + 60RE + 5R^2 \mathbf{1}) = \frac{R^2}{72}$$

(the R^2 /Starobinsky term), so the spectral action structurally produces $S_{\text{eff}} = \int \sqrt{g} \frac{\bar{M}_{\text{Pl}}^2}{2} (R + \frac{R^2}{6\bar{M}^2}) + \dots$ with the scalaron mass ratio fixed by the cutoff moment, $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$. The TFPT closure $M^2/\bar{M}_{\text{Pl}}^2 = c_3^7$ fixes $f_0 = 6(4\pi)^2/c_3^7$, i.e. $M = c_3^{7/2} \bar{M}_{\text{Pl}} = 3.06 \times 10^{13} \text{ GeV}$ — the boundary normalisation c_3 is the gravitational scale — and hence the Starobinsky slow-roll forms $n_s \simeq 1 - \frac{2}{N}$, $r \simeq \frac{12}{N^2}$. The $R + R^2$ structure is convention-independent (standard Gilkey); the precise rational $\frac{1}{72}$ and factor 6 are the standard Dirac conventions used here ([`verification/v36_spectral_action_g2.py`], [`verification/v28_gravity_fR.py`]).

The simpler resolution: “full QG” for physics is the gap-decoupled admissible sector ([E]). The infinite-dimensional ambient measure (G6) is the summit, but the admissible IR sector does not wait for it. The metric coupling is gap-dominated (G5): $2\|V_{\text{metric}}\| = 0.785 < \Delta = 6 \log \frac{3}{2} = 2.433$, so the admissible sector keeps a *strictly positive* effective gap $\Delta_{\text{eff}} = 1.648 > 0$ under the stated RP and metric-coupling assumptions. A positive gap dynamically *isolates* the admissible sector from the un-built ambient/deep-UV: under those assumptions the admissible (IR) measure is closed, and the ambient completion is decoupled from it (we do *not* assert it is physically irrelevant — that would itself be a physical interpretation). So the honest reading is

admissible IR sector = closed under stated RP; ambient metric measure (G6) = G_{metric} gate
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G2 supplies the local action ($R + R^2$, heat-kernel); G5 supplies the decoupling of the admissible sector; together they reduce (G_{metric}) to the ambient projective limit (G6). **Precise status (Alessandro’s boundary):** the admissible IR sector is closed under the stated RP and gap-dominance assumptions; the ambient metric projective measure remains a *strict TOE completion target*. Its absence does *not* affect the bounded IR claim, but it *does block certification as a strict physical TOE* — so we do not call it “mere mathematical completeness”. [`verification/v36_spectral_action_g2.py`]

The glue Q-system: the index-4 object of the G_{net} statement exists explicitly ([E]/[C]). The G_{net} target — “the seam–Calderón inclusion has Jones index $4 = |\mu_4|$ ” — no longer refers to an unknown object. On the glue group algebra $A = \mathbb{C}[\mathbb{Z}_4]$ (multiplication $\delta_a \otimes \delta_b \mapsto \delta_{a+b}$, unit δ_0) all Longo Q-system axioms hold exactly [E]: associativity, unit law, the Frobenius identity $(m \otimes 1)(1 \otimes m^*) = m^* m = (1 \otimes m)(m^* \otimes 1)$, and specialness $mm^* = |\mathbb{Z}_4| \mathbf{1}$ — a special C^* Frobenius algebra with

$\dim \theta = 4 = \mu_4 = [B : A]$

exactly the Carrier Index Lemma value (KLM: $\mu_{\text{carrier}} = 16 = 4^2 \mu_{E_8}$ re-verified). *Locality = isotropy* [E]: on the Lagrangian glue $\langle (1, 1) \rangle$ the discriminant form gives $q(k(1, 1)) = k^2 \in \mathbb{Z}$ for every k — the braiding obstruction vanishes (v92 in Q-system language); the two Lagrangian glues (the sheet pair) are the only maximal isotropic choices. *Halfway step* [E]: $\{0, 2\} < \mathbb{Z}_4$ closes and gives the unique intermediate Q-system $\mathbb{C}[\mathbb{Z}_2]$ = the $SO(16)_1$ tower step, with index factorisation $4 = 2 \times 2$. *R2 sharpened* [C] (recorded, not claimed): what remains is the operator-algebraic *identification* “the seam–Calderón inclusion of the 16-Majorana net (v113) realises *this* Q-system” — matching two constructed objects, not a search; the three loads (metric + carrier + QBL, v123) hang on this one identification.

The hull is the graded module of the Q-system ([`verification/v128_graded_hull.py`]). The Q-system is not merely an abstract companion of E_8 — E_8 realises it [E]: the 240 roots, built as

explicit norm-2 vectors in $D_5 \oplus A_3$ +glue coordinates, decompose as $240 = 52 + 64 + 60 + 64$ over the glue $\mathbb{Z}_4$ (the v89 branching as vectors), and for *all* 6720 root pairs whose sum is again a root, $\text{coset}(r_1+r_2) = \text{coset}(r_1) + \text{coset}(r_2) \bmod 4$:

E_8 is a $\mathbb{Z}_4$ -graded Lie algebra over the carrier; the grading group is the glue = the Q-system

— the checksum hull is the Q-system’s graded module, realised in 248 dimensions. The R2 identification thus matches the Calderón inclusion against a structure that E_8 already *carries*.
[verification/v125_glue_qsystem.py]

Graded Frobenius ([verification/v143_graded_frobenius.py]). The identification holds at the finite Lie level: the graded module is in fact a graded *Frobenius* structure, i.e. everything the Q-system demands of its module is already exact inside E_8 [E]: (i) *coset duality* — $\text{coset}(-r) = -\text{coset}(r) \bmod 4$ for all 240 roots ($-C_1 = C_3$ bijectively, C_0, C_2 self-dual), so the Killing form restricts to a *nondegenerate* invariant pairing $g_k \times g_{-k} \rightarrow \mathbb{C}$; (ii) the glue character average $\frac{1}{4} \sum_k i^{k \deg}$ is *exactly* the carrier projector (the conditional-expectation shadow, quasi-basis index $|\mathbb{Z}_4| = 4$); (iii) each glue sector is *one* $W(D_5) \times W(A_3)$ orbit with multiplicity-one dimensions $64 = 16 \cdot 4 = (\mathbf{16}, \mathbf{4})$, $60 = 10 \cdot 6 = (\mathbf{10}, \mathbf{6})$ — simple currents, sector ring $= \mathbb{Z}_4 =$ the $\mathbb{C}[\mathbb{Z}_4]$ fusion, halfway subgroup $\{0, 2\} = SO(16)_1$. *R2 residue [C] (final form)*: exactly *one* conformal-net statement remains — the seam–Calderón inclusion of the 16-Majorana *net* carries this Q-system — with no remaining finite freedom.

Where that statement lives: the NS/R sector census ([verification/v148_fock_census.py]). The remaining work is scoped exactly. Every *untwisted* (Neveu–Schwarz) state of the 16 Majoranas has *integer* ($D_5, SO(6)$) weights, and integer weights lie in the even discriminant classes — so the NS module supports only $\{(0, 0), (2, 0), (0, 2), (2, 2)\}$: **the odd glue classes $k(1, 1)$ — the actual μ_4 simple currents — have zero NS support; they are twisted (Ramond-type) modules.** The Ramond zero-mode module ($2^8 = 256$ ground states) carries exactly the four odd class pairs, 64 each, and splits $128 + 128$ into the odd sectors of the *two* Lagrangian glues $\langle(1, 1)\rangle$ and $\langle(1, 3)\rangle$ (v92): *choosing the glue is choosing the Ramond projection* ($128 =$ one $SO(16)$ spinor chirality), and E_8 assembles as $248 = 120_{\text{NS}} + 128_{\text{R}}$. *Consequence [C]*: the index-4 extension cannot even be *stated* on the untwisted module alone; R2’s one remaining statement is irreducibly about the twisted-sector structure of the seam–Calderón net, with the sheet choice realised as the R-projection; the finite shadow is exhausted (v113/v125/v143 + this census).

Lemma 10 (G3–G4 — reflection positivity, fixed and integrated). *For each Θ -invariant g in the Calderón ball the physical boundary kernel $K_g = P_{\text{adm}} \Pi_{\text{phys}} \mathcal{N}_g^{TT} \Pi_{\text{phys}} P_{\text{adm}}$ is reflection positive, $\langle \Theta F, F \rangle_{K_g} \geq 0$ (G3); and $d\mu_{\text{Cal}, \chi}$ is Θ -invariant, supported where G3 holds uniformly, so $\int \langle \Theta F, F \rangle_{K_g} d\mu_{\text{Cal}, \chi} \geq 0$ (G4). **G4 is the decisive step:** RP after integrating over dynamical metrics, not merely at fixed background.*

Lemma 11 (G5 — gap dominance [E]). $2\|V_{\text{metric}}\|_{\text{rel}} < \Delta = 6 \log \frac{3}{2}$, with $\|V_{\text{metric}}\|_{\text{rel}} \leq 248c_3^2 = \frac{31}{8\pi^2} = 0.39262$, hence $2\|V\| = 0.7852 < 2.4328$ and $\Delta_{\text{eff}} \geq \Delta - 2\|V\| > 1.647 > 0$. *The numerical inequality is trivial ([verification/v29_research_contract_certs.py]); the work is the operator-norm bound (hard).*

Sugawara reading + atomic OS moment ([verification/v171_os_moment_cluster.py]). The G5 budget is the E_8 Sugawara constant: $248c_3^2 = \frac{31}{8\pi^2}$ with $31 = 1 + h^\vee(E_8)$ and $c((E_8)_1) = 248/31 = 8$, so gap dominance reads $|R^+(A_3)| \log(N_{\text{fam}}/|\mathbb{Z}_2|) > \frac{1+h^\vee(E_8)}{4\pi^2}$ — the A_3 transfer gap beats the E_8 Sugawara metric budget. Independently, the G3 reflection positivity is a *positive moment problem*: $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ makes every two-point correlator $C(n) = A + B(64/729)^n + C(1/729)^n$ the moment sequence of a positive atomic measure, so each OS Hankel matrix factorises as a Vandermonde Gram and is positive semidefinite (a clean RP certificate; cluster constant $729/665$, correlation length $\xi = 1/(6 \log \frac{3}{2}) = 0.4111$).

Theorem 1 (Decoupling theorem — physical low-energy claims do *not* need the full G_{metric}). If the admissible sector is reflection-positive (G_4) and the metric coupling is gap-dominated, $2\|V_{\text{metric}}\| < \Delta = 6 \log \frac{3}{2}$ (G_5), then the admissible IR measure is stable under the ambient completion: a strictly positive effective gap $\Delta_{\text{eff}} = \Delta - 2\|V\| > 0$ protects the admissible sector from the (un-built) ambient/deep-UV, so every physical low-energy read-out (masses, mixings, α^{-1} , the $R+R^2$ regime) is independent of the open ambient measure. Symbolically

$$\text{RP} + 2\|V\| < \Delta \implies \text{admissible IR measure stable under ambient completion.}$$

Thus (G_{metric}) is not a diffuse “full quantum gravity missing” block: it is the sharply delimited ambient projective limit (G_6), a strict-TOE certification target whose closure the falsifiable IR physics does not depend on. **Typing:** the inequality is machine-checked [E]; the operator-norm bound on $\|V_{\text{metric}}\|$ is the conditional input.

Lemma 12 (G_6 – G_7 — projective limit & Ward identities). A projective system μ_{χ_n} with $\pi_{m,n*}\mu_{\chi_m} = \mu_{\chi_n}$, uniform moment bounds $\sup_n \int \|g\|_{H^s}^p d\mu_{\chi_n} < \infty$, tightness and regulator-independence yields $\mu_{\text{QG}} = \lim \mu_{\chi_n}$ on $\mathcal{G}/\text{Diff}$ (G_6 , the hardest step); the limit obeys the BRST Ward identities $\langle s\mathcal{O} \rangle = 0$, whence $\nabla^\mu T_{\mu\nu} = 0$ and the η boundary phase cancels the APS anomalies (G_7).

Gate 2 reduction: G_6 is a *boundary projective limit*, not a bulk one ([verification/v76_gmetric_reduction.py])

The exact decoupling margin is $\Delta_{\text{eff}} = \Delta - 2\|V\| = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$, with $\|V_{\text{metric}}\|_{\text{rel}} \leq 248 c_3^2 = \dim(E_8) c_3^2 = \frac{31}{8\pi^2} (31 = 2^{g_{\text{car}}} - 1)$. **Holographic reduction:** because the seam is a *finite causal boundary*, the ambient *bulk* metric measure is reconstructed from a finite-codimension seam (Calderón) *boundary* measure — so G_6 is a *boundary projective limit*. The conditional theorem is then

$$\text{RP}(\text{seam-Calderón boundary kernel}) + \text{tightness} \implies G_{\text{metric}} \text{ closes}$$

i.e. the dimension of the open problem drops from bulk QG to a finite seam-boundary measure. **Residual:** the constructive boundary projective limit (constructive-QFT grade) — reduced, not closed. So Gate 2 is now *two clean tiers*: Tier A (IR physics) closed under RP+gap; Tier B (strict TOE) reduced to a boundary measure, no longer a diffuse “full QG missing”. [C]/[O]

And that boundary measure is a rigorously constructed net ([verification/v77_e8_conformal_net.py]). The level-1 central charges $c = \dim \mathfrak{g}/(1 + h^\vee)$ give $c(E_8)_1 = \frac{248}{31} = 8 = \text{rank } E_8$, $c(D_5)_1 = \frac{45}{9} = 5$, $c(A_3)_1 = \frac{15}{5} = 3$, and $c(D_5) + c(A_3) = 5 + 3 = 8 = c(E_8)$ is a *conformal embedding* $(D_5)_1 \times (A_3)_1 \subset (E_8)_1$ (coset $c = 0$). Now $(E_8)_1$ is the holomorphic $c = 8$ E_8 -lattice VOA / conformal net — one of the most rigorously constructed objects in mathematical QFT (Frenkel–Kac–Segal; Kawahigashi–Longo / Carpi). **So the conditional theorem becomes concrete:** identify the seam-Calderón boundary measure with the $(E_8)_1$ lattice net (carrier = $(D_5)_1 \times (A_3)_1$ subnet), and the gap $\Delta_{\text{eff}} > 0$ supplies the clustering/tightness — so G_6 is imported into existing RCFT/conformal-net rigor rather than built from scratch. Residual = the net identification + bulk reconstruction. [E] (charges, embedding) + [C] (identification)

The Simple-Current Extension Theorem — the central G_{net} closing statement [E]/[C] ([verification/v154_simple_current_theorem.py])

The conditional theorem can now be stated as one clean *simple-current extension*. Let $A = (D_5)_1 \otimes (A_3)_1$ be the carrier net and $L = \langle (1, 1) \rangle \subset \text{Disc}(D_5) \oplus \text{Disc}(A_3) = \mathbb{Z}_4 \oplus \mathbb{Z}_4$ the isotropic μ_4 glue line ($q(k(1, 1)) = k^2 \in \mathbb{Z}$). Then the local extension $B = A \rtimes L$ has

$$[B : A] = |L| = 4 = |\mu_4|, \quad c(B) = 5 + 3 = 8, \quad \mu(B) = \frac{\mu(A)}{|L|^2} = \frac{16}{16} = 1$$

so B is *holomorphic* (KLM μ -additivity), and a holomorphic $c=8$ chiral net is the lattice net of the *unique* even-unimodular rank-8 lattice — hence $B \cong (E_8)_1$ (the same- c rival $SO(16)_1$, $\mu=4$, is excluded). Every algebraic ingredient is machine-checked: the $\mathbb{C}[\mathbb{Z}_4]$ Longo Q-system ([verification/v125_glue_qsystem.py]), the $\mathbb{Z}_4$ -graded Frobe-

nius E_8 hull ([verification/v143_graded_frobenius.py]), the Ramond glue sectors ([verification/v148_fock_census.py]). **Consequence:** G_{net} is internally *closed* if the seam-Calderón boundary net is *defined* to be A ; it stays the *one* external identification “the RP seam-Calderón net is A ” if the boundary net is to be *derived* from the seam kernel. Either way this is the single strict-TOE certification theorem still carrying weight — a clean extension statement, not “find a quantum gravity”.

And that one identification reduces to a single shared premise ([verification/v155_quasifree_boundary.py]). The identification splits (v110) into (a) “the seam Calderón involution is sheet-odd” — the double-cover deck, the same $|\mathbb{Z}_2|$ that halves $c_3 = \frac{1}{2 \cdot 4\pi}$ (structural) — and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not independent*: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression PTP of the bulk covariance, and the compression of a contraction ($i\Gamma$ with spectrum in $[-1, 1]$) is a contraction — so the boundary/seam state is automatically quasi-free, with Wick inherited (every boundary correlator a Pfaffian of Γ_d). Hence (b) *follows* from “the seam bulk is a reflection-positive free field” — the *same* premise already load-bearing in the area law ([verification/v59_area_law_evidence.py]) and the EH mechanism ([verification/v150_replica_eh_model.py]/[verification/v151_bfk_split.py]). **So the entire seam-side residual is one physical premise** — the seam bulk is a free RP field — shared by G_{net} , the area law and R3, and given it the chain Gaussian bulk $\Rightarrow$ quasi-free boundary $\Rightarrow$ one kernel = net (v113) $\Rightarrow$ carrier $A \Rightarrow (E_8)_1$ is exact. It is not reducible by a finite computation — it is the one place TFPT meets functional analysis.

The construction, exhibited and verified ([verification/v156_seam_net_construction.py]). The “ $B \cong (E_8)_1$ ” step is no longer an assertion — the chiral net is built explicitly and the identification *checked*. The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian acts on a boundary mode e^{ikx} by the harmonic extension $e^{ikx-|k|y}$: $-\partial_n u|_0 = |k|u$, so for a free bulk the seam operator is the massless chiral dispersion $\omega_k = |k|$ that defines the free-fermion net on S^1 . Sixteen Majoranas give $c = 8 = 5+3$; the weight-1 currents are $248 = 120_{\text{NS}} + 128_{\text{R}}$ (the $SO(16)$ bilinears plus the Ramond spinor, Frenkel-Kac); and the holomorphic character is

$$\chi = \frac{E_4}{\eta^8} = q^{-1/3} (1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + \dots) = j^{1/3}$$

(verified by q -series), the 248 at level 1 being $\dim E_8$ — so the net *is* $(E_8)_1$. *What remains* is exactly (i) the free-bulk premise above and (ii) the operator-algebraic existence / uniqueness of the net generated by the seam Calderón kernel (a Kawahigashi-Longo-type net-construction theorem). Route 3 thus *exhibits and verifies* the target; the irreducible residual is the one premise + the standard net-existence step — where TFPT meets constructive QFT, now sharply located.

And the free-bulk premise itself reframes: freeness is a rigid fixed point, not a postulate ([verification/v157_rigid_fixed_point.py]). Three facts click together. (i) The DtN/Calderón symbol is *universally* $|k|$: the boundary operator of *any* second-order elliptic bulk is an order-1 Ψ DO with principal symbol $|\xi|_g$ (Lee-Uhlmann), so the free chiral dispersion is bulk-detail-independent and any interaction lives in the lower-order (irrelevant) symbol. (ii) A holomorphic $c=8$ theory is *rigid*: every field has weight $(h, \bar{h}=0)$, an exactly-marginal deformation needs $(1, 1)$, and with $\bar{h}=0$ *no* $(1, 1)$ field exists — the fixed point is isolated, there is no continuous knob along which a non-Gaussian interaction could turn on. (iii) The *same* $|\mathbb{Z}_2|$ that halves the one-sided S^2 Gauss-Bonnet ($c_3 = \frac{1}{|\mathbb{Z}_2| 2\pi \chi(S^2)} = \frac{1}{8\pi}$) labels the sheet *and* is the Ramond/GSO projection giving $\mu=1$ ($248=120_{\text{NS}}+128_{\text{R}}$) — one object, three roles, so the geometry fixing c_3 is the projection making the net holomorphic. **Consequence:** premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ($\Delta=6 \log \frac{3}{2} > 0$) boundary flow reaches the holomorphic fixed point” — given which, freeness is *forced* by rigidity + uniqueness ($(E_8)_1$, v83), not fine-tuned. A strictly smaller and simpler residual.

And the destination is a proven stable fixed point ([verification/v158_fixed_point_stable.py]). An exact operator-dimension count settles it: the chiral Majorana has $h=\frac{1}{2}$, the bilinear current $h=1$, the quartic $h=2$. The relevant window $0 < h < 1$ for bosonic deformations is *empty* (bosonic chiral operators have integer h , lowest 1) — nothing relevant flows the free theory away; the $h=1$ currents are chiral (v157), not $(1, 1)$ -marginal; and the lowest genuinely-interacting operator (the quartic, $h=2>1$) is *irrelevant*, as are all higher $2k$ -fermi operators ($h=k$). So the free chiral $c=8$ fixed point is *isolated and stable* against all interactions. Premise (A) thereby reduces to a *basin-of-attraction* statement with a proven stable target — the gapped ($\Delta=6 \log \frac{3}{2}$) flow lands here; the convergence is conformal-perturbation analysis, the destination is settled.

And the closure can be stated as one fixed-point theorem — which relocates (A) to a single internal premise ([verification/v160_seam_gaussianity_from_pf.py]). Phrase the argument internally: (i) a quasi-free state has *all* higher connected cumulants $\kappa_{2n}=0$ (the signed set-partition/Pfaffian inversion of $\langle\psi_1\cdots\psi_{2n}\rangle = \text{Pf}(\Gamma)$ leaves only the degree-2 kernel — the v113 “one kernel = the net” property at the cumulant level); (ii) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (iii) a gapped primitive Perron–Frobenius transport has a *unique* fixed ray, so a non-Gaussian piece is impossible — if κ_{2n} were fixed it would be a *second* fixed ray (killing uniqueness), and if not fixed it sits in the gapped complement and decays to 0 (any gap > 0 suffices; the value $(2/3)^6$ only sets the rate). *The honest residual:* v56’s gapped PF uniqueness is established on the *three-dimensional* scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (the flavour gap / horizon Page-recovery rates), whereas the 16-Majorana seam state already carries $\binom{16}{4}=1820$ and $\binom{16}{6}=8008$ higher-cumulant directions — the Schwinger cone the transport must contract has dimension $\gg 3$, and uniqueness must hold there (not merely on the quasi-free sub-cone) for the argument to be non-circular. **So (A) is equivalent to one strictly smaller, internal, exact-testable statement:** the admissible TFPT seam transport is a *primitive, spectrally gapped, RP-cone-preserving* map on the *full* Schwinger cone, not only on the scalar readout spectrum of v56 — which the suite does *not* yet establish [E]/[O].

And that full-cone premise is now *discharged* to a single finite one-particle statement ([verification/v161_one_particle_reduction.py]). The lever is Bogoliubov second quantisation: a quasi-free transport is $T_\Sigma = \Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dimensional Majorana space H_Σ , and the Fock space grades by correlator degree, so $\Gamma(t)$ acts on the degree- m sector as the exterior power $\Lambda^m(t)$. Three exact consequences follow (all machine-checked): (i) $\text{spec}(\Gamma(t)|_{\deg m})$ is the set of products of m one-particle eigenvalues — the whole correlation spectrum is generated by the finite one-particle spectrum; (ii) **the cone gap equals the one-particle gap:** the sub-leading eigenvalue of $\Gamma(t)$ over *every* degree is $(2/3)^6$ (pad the largest contracted mode with fixed factors), so premise 5 on the cumulant space *is* premise 5 on the 16-dim space, the same number; (iii) the eigenvalue-1 sector is Λ^* of the fixed subspace — the disconnected Wick powers of K_Σ — so with a strict one-particle gap no independent connected cumulant is fixed, and an independent fixed cumulant *on* the fixed subspace is excluded by v113 (one polarisation $\Leftrightarrow$ one vacuum). Premise 1 (quasi-freeness) is itself *forced*, not assumed: the $120 = \dim \mathfrak{so}(16)$ weight-1 chiral currents *are* the Bogoliubov generators on the 16 Majoranas, and v158 makes every non-Gaussian deformation irrelevant. **So the three full-cone premises collapse to one finite, 16-dimensional statement:** the seam RG transport’s one-particle symbol is the gapped Calderón Bogoliubov map — fixed eigenspace = the rank-8 K_Σ polarisation (v113), sub-leading gap = $(2/3)^6$ (v56). Every *number* in it is already supplied by the suite; the one irreducible step is the physical identification that the actual seam RG generator equals this Bogoliubov map — the same operator-algebraic (Kawahigashi–Longo-class) statement as the net-existence residual of (A₂), now with the infinite-dimensional cone eliminated [E]/[O].

And that finite statement carries no *new* open content: it factors into the two already-open items ([verification/v162_seam_transport_identification.py]). Take the one-particle symbol apart into its two data, the gap value (β) and the fixed space (α). (β) *is forced*: a μ_4 -equivariant operator is diagonal in the cusp basis — the deck average is the diagonal projection, $\sum_{k=0}^3 U^k X U^{-k} = |\mu_4| \text{diag}(X)$ for $U = \text{diag}(1, i, -i)$, and the commutant of U is exactly the diagonal algebra — so its spectrum is the cusp weights $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (the parabolic residue, v115), and the transfer eigenvalue $(1-\alpha)^{p_2}$ with $p_2=6=|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6 \log \frac{3}{2}$ — every entry carrier data, no free knob; the lift to the 16 Majoranas is the established A_3 subnet grading (v137). (α) is the rank-8 Calderón polarisation of the $\omega_k=|k|$ free dispersion (v113/v156), the IR conformal free-fermion vacuum. **What is left** is only that the seam RG generator *is* μ_4 -equivariant and admissible — which is the R1 “seam clock” identification (the transfer operator as the seam’s gapped generator) — together with the A_2 net-existence theorem. Both are pre-existing open items, so premise (A) introduces *no independent open content*: it factors exactly into $A_2 + R_1$ with every number forced. The “free-bulk cloud” is gone; what remains is the same two identifications the programme already names — (A) is closed *conditionally* on them (a unification, honestly not an unconditional proof) [E]/[O].

And both residual identifications are pinned to their floor, so (A) has no independent open gate left ([verification/v165_premise_a_closure.py]). A_2 (net existence) is not open research but *assembly* of established mathematics: the 16 free Majoranas define a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic $c=8$ lattice net *is* $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125); the one TFPT-specific input — that the seam Calderón kernel is the free-fermion two-point function — is done because the DtN symbol is universally $|k|$ (v156/v157). So A_2 re-types [O] $\rightarrow$ [C] (closed by assembly). R_1 (the

seam clock) reduces to **GATE.QGEO**: a μ_4 -equivariant flat gapped generator on the carrier is *unique* ($= A_0^*$, [v115/v116](#)), so its gap $(2/3)^6$ is forced once the seam generator lies in that class, and the residual is exactly one geometric identification — the seam collar boundary is the μ_4 -punctured sphere $\mathbb{P}^1 \setminus \mu_4$ — whose cohomological half is already established ([v137](#), $b_1=3=N_{\text{fam}}$). That is **GATE.QGEO**, an already-open realisation premise, *not* a new gate. By the No-Unit Theorem ([v153](#)) the irreducibles stay exactly $\{\pi, v_{\text{geo}}\}$ — closing (A) adds none. **Final accounting:** premise (A) = (A_2 assembly, [\[C\]](#)) + (R_1 =**GATE.QGEO**, already open) + the $\{\pi, v_{\text{geo}}\}$ irreducible core (a theorem) — it carries *zero* independent open gates and ceases to be its own gate [\[E\]/\[O\]](#).

And the two structural residuals — net existence and full-cone reflection positivity — are now discharged to a theorem, leaving the seam realisation as the one irreducible open premise ([\[verification/v175_net_existence_full_cone.py\]](#)). The CAR second-quantisation functor $\Gamma(t) = \bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the *complete* Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16}=65536$); by the exterior-power spectrum theorem every eigenvalue of $\Gamma(t)$ is a subset product of $\text{spec}(t)$, so checking all 2^{16} products shows $\Gamma(t)$ is a *contraction* (full-cone RP), with eigenvalue-1 multiplicity $2^8=256$ (the wedges of the rank-8 K_Σ polarisation) and sub-leading eigenvalue exactly the one-particle gap $(2/3)^6$ — so full-cone RP/gap reduce *for every* m (not only $m \leq 6$, [v161](#)) to the one-particle contraction (Shale–Stinespring), *discharging the v160 scope premise by a theorem*. And A_2 is now an *assembled and verified* certificate: the $(E_8)_1$ character $E_4/\eta^8=1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$ (level-1=248), the embedding $248=120+128$, and the E_8 Cartan matrix even with $\det 1$ (the unique rank-8 even unimodular lattice, [v83](#)) — the net *exists* by cited theorems. **Both residuals then collapse to the same input:** that the actual seam Calderón kernel is this gapped one-particle contraction on $\mathbb{P}^1 \setminus \mu_4$, i.e. **QGEO.REALIZE.01**, whose finite half is proven ([v168](#)). That realisation half is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [\[O\]](#) — the single irreducible structural premise; net existence and full-cone RP are [\[E\]](#).

The No-Unit Theorem — v_{geo} is an irreducible metrology primitive [\[E\]/\[O\]](#) ([\[verification/v153_no_unit_theorem.py\]](#))

v_{geo} is not an open gap. Every load-bearing compiler datum — $g_{\text{car}}, |\mu_4|, N_{\text{fam}}, \text{rank } E_8, \alpha^{-1}$, the determinant ladder $(3, 4, 8, 20)$, $c_3 = \frac{1}{8\pi}$, φ_0^{ret} — is mass-dimension 0 and *invariant* under a unit rescaling $L \mapsto \lambda L$, whereas a mass / energy / $1/G$ carries nonzero dimension. A map from dimension-0 inputs to a dimension- $d \neq 0$ output is simultaneously invariant and covariant: impossible unless the unit is introduced. **So a dimensionless boundary compiler provably cannot select an absolute scale** — v_{geo} is a *forced input*. The three apparent residuals then collapse onto the one unit, differing only in mass-dimension:

$$U_{\text{point}} \sim v_{\text{geo}}, \quad \frac{1}{G} \sim v_{\text{geo}}^2, \quad m/\mu = e^{3/4} \text{ (a pure ratio),}$$

so the irreducibles stay exactly {one dimensionful unit, π }. *Re-typing:* v_{geo} is an *irreducible metrology primitive* (a declared unit, like choosing a metre), not a missing derivation.

Theorem G — the closure statement [\[O\]](#) (target)

The relative spectral boundary kernel over the diffeomorphism-quotiented metric sector admits a regulator-independent, reflection-positive projective-limit measure μ_{QG} . Its Ward identities imply covariant conservation, and its low-curvature expansion is the TFPT $R + R^2$ seam action

$$f_R R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu} \square - \nabla_\mu \nabla_\nu) f_R = \bar{M}_{\text{Pl}}^{-2} T_{\mu\nu} + O(R^3/M^4),$$

so the closed covariant equation becomes the controlled IR shadow of the full measure. *Status: open; the deepest item. G5 certified; the regime equation closed* [\[verification/v28_gravity_fR.py\]](#).

Certifiability and recommended order

Building block	Machine	Hardness
W_{wall} enumeration ($C_U^{(1)}$)	Lean / Python / Wolfram	easy (<i>done</i>)
D_4 fixed-locus polynomial ($C_U^{(2)}$)	Sage / Mathematica	medium
selector uniqueness ($C_U^{(3)}$)	interval arithmetic	medium
SNF, determinants, spectra	Lean	easy–medium
principal-symbol ellipticity (G1)	xAct / Cadabra / Lean	medium
gap inequality (G5)	Lean	easy (<i>done</i>)
operator-norm bound (G5)	analysis + numerics	hard
projective limit (G6)	not primarily machine	very hard
Ward identities (G7)	BV/BRST formal + checks	hard

Recommended order.

Selector-triangle pairings $\rightarrow v_{\text{geo}}$ scale anchor $\rightarrow G_{\text{net}}$ inclusion theorem

The flavor interface (historically U_{wall}) is now reduced to the *selector-triangle pairings* (the dual normal pair (d, n) forcing R columnwise, v136/v139); it is finite, falsifiable and the cheapest visible upgrade. The v_{geo} scale anchor is the one dimensionful input (same nature as $1/G$). G_{net} — the index-4 seam-net inclusion certifying the metric sector — is more fundamental but a *programme*, not a single proof: heavy machinery, with the projective limit the genuine summit. F_{transfer} (source→pole / relic / cosmology) is the remaining downstream interface.